

CHAPTER 4

Architecture Definition



This chapter is part of the PINK and GREEN reading tours. See Figure 1-1 in Chapter 1, Section 1.3, for alternative reading tours.

Definitions are of the utmost importance in creating clarity about what is being talked about or what is being dealt with. A definition provides an unambiguous description of the subject, leaving no room for interpretation. However, when this freedom of interpretation is provided, it can lead to different perceptions of the subject, with all the consequences that entails.

Looking at the field of Enterprise Architecture, it is important to have an unambiguous definition. In this chapter, the many definitions of Enterprise Architecture are presented. It is also pointed out that some essential parts are missing.

4.1. What Is Enterprise Architecture

One of the topics that has been debated over the years is what exactly is meant by architecture (or Enterprise Architecture, for that matter). In the current version of the TOGAF Standard, the following definition is given:

The fundamental concepts or properties of an entity in its environment and governing principles for the realization and evolution of this entity and its related lifecycle processes [7].

With the publication of the Technical Corrigendum 1 (2025), the TOGAF Standard currently adopts the ISO/IEC/IEEE 42010:2022 [7] standard for defining Enterprise Architecture. The given definition still leaves room for interpretation. To fill this gap, the framework has added additional text to the above, a sort of second definition:

The structure of components, their interrelationships, and the principles and guidelines governing their design and evolution over time [2].

Both definitions - or rather the combination of the two definitions - indicate that architecture defines the characteristics of a system and that the system is surrounded by frameworks that oversee its development. According to the TOGAF Standard, the word *system* can mean several things and is not directly related to an IT system. The word *system* can just as easily refer to an organization (an enterprise) or be even larger in nature. On the other hand, it can just as easily refer to a project. So there is plenty of room for interpretation.

Chris Greenslade, who was Chairman of The Open Group in 2003, gave a presentation on TOGAF 7, including a look ahead to version 8. There were some interesting things mentioned in the presentation that can be considered progressive, especially considering the time they were in at that point. More about this is described in Chapter 2, Section 2.2.

The presentation included a definition of what was understood to be architecture in the year 2003:

“Conceptually an (IT) Architecture is the fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution.

Practically it is represented in Architectural Descriptions from the viewpoints of the Stakeholders” [8].

What is surprising about today’s definition of Enterprise Architecture in the TOGAF Standard is that it is missing an important component: its relation to the achievement of business objectives. Enterprise Architecture must ensure that these objectives and their intended results are achieved, and it must shape and organize itself to make this possible. In doing so, Enterprise Architecture must ensure consistency and coherence between concepts (from organizational structures to information systems and from capabilities to applications).

Enterprise Architecture is perhaps best described as a framework for understanding and managing the overall structure and strategy of an organization.

It is about creating a holistic view of the organization's activities, including its business processes, information systems, and technology infrastructure, and maintaining the coherence between them.

The purpose of Enterprise Architecture is to align these various elements with the goals and objectives of the organization and to ensure that the elements work together effectively and efficiently.

Enterprise Architecture focuses on identifying and resolving inconsistencies in business operations and enables planning for future growth and development.

Definitions are very important because they provide clarity about what is being talked about or what is being dealt with. A definition provides an unambiguous description of the subject, thus avoiding confusion.



Further Reading

Chris Greenslade's 2003 presentation can be found at this URL:

https://www.opengroup.org/architecture/0307bos/presents/greenslade_togaf.pdf

4.2. Confusion Over Definitions

Enterprise Architecture is still often confused with IT Architecture. As was made clear in Chapter 2, Section 2.2, Enterprise Architecture has its origins in the technology corner. Perhaps part of the problem is that Enterprise Architecture is still lumped together with IT Architecture. In reality, the two are as different as night and day.

Enterprise Architecture is first and foremost about enabling an organization to achieve its goals and objectives, thereby realizing the organization's strategy. It is used to address stakeholder concerns and answer any questions they may have regarding the organization's portfolio or specific projects.

IT Architecture, on the other hand, is far more technical in nature compared with Enterprise Architecture. Therefore, the two are miles apart from being the same. There is absolutely *nothing technical* to Enterprise Architecture. The diagrams and matrices used in architecture efforts are often being referred to as being technical in nature. However, when used in an Enterprise Architecture practice, they answer stakeholders' questions and address their concerns. They do not illustrate technical solutions.

Enterprise Architecture oversees all the architecture domains (of which Technology Architecture is one) and plays a coordinating role. It leaves the execution of Technology Architecture to the IT Architects.

As such, Enterprise Architecture is considered a strategic business management tool, not a technical instrument [9].

The definition of Enterprise Architecture used by the TOGAF Standard is only one of many that exist. No single definition can be considered the ultimate one. Perhaps that is why even the TOGAF Standard uses more than one definition of Enterprise Architecture.

4.3. Multiple Definitions

As mentioned earlier, the definition shown at the beginning of this chapter is the one included in the Core Concepts Guide of the TOGAF Standard. The definition is based on the ISO/IEC/IEEE 42010:2022 standard, which is maintained and updated by ISO.

Chapter 3 attempted to paint a picture of the composition of the TOGAF Standard. The section called Fundamental Content was explained, as well as the additional Series Guides. One of these Series Guides, the Leader's Guide to be precise, describes a second definition of Enterprise Architecture:

The process of translating business vision and strategy into effective enterprise change by creating, communicating, and improving the key principles and models that describe the enterprise's future state and enable its evolution [3].

This definition was originally provided by Gartner [10] and was created in 2008, *years* before the 2022 ISO definition. Given the speed with which the profession evolves (and has evolved) over the years, it's safe to say that using a definition drawn up in 2008 in an architecture framework updated in 2025 might raise a few eyebrows. Not to mention the fact that there is a real chance that this definition is outdated and superseded.

Figure 4-1 shows the collection of all Enterprise Architecture definitions used within the TOGAF Standard or referenced directly or indirectly through the use of other bodies of knowledge.

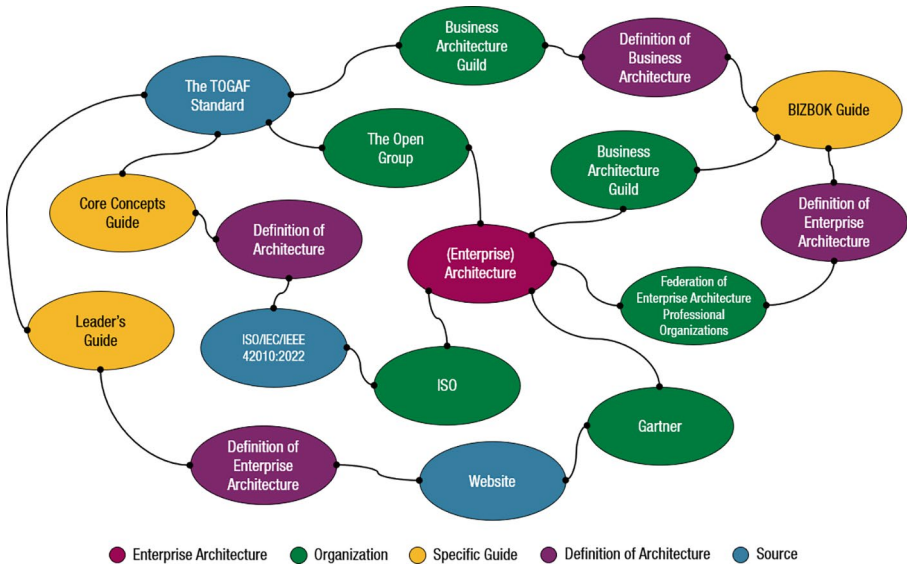


Figure 4-1. *The many definitions of Enterprise Architecture*

Figure 4-1 can be read as follows. At the center is Enterprise Architecture as the primary object to be defined. Immediately around it are The Open Group, the Business Architecture Guild, the Federation of Enterprise Architecture Professional Organizations (FEAPO), Gartner, and ISO. All are organizations that either provide their own definition of Enterprise Architecture or played a significant role in formulating the definition as ultimately adopted by the TOGAF Standard.

The various sources from which the individual definitions can be found are indicated in blue. For example, the definition of the profession as described in the TOGAF Standard can be found in the tenth edition of the framework. Gartner, for example, has published the definition of Enterprise Architecture on its website.

The yellow ovals indicate the specific references to the sources in which the definitions are articulated. Finally, the purple ovals indicate a definition of either Enterprise Architecture, Business Architecture, or Architecture.

One of the bodies of knowledge referenced in the TOGAF Standard is the BIZBOK Guide. This body of knowledge deals with the concepts within the domain of Business Architecture and goes into considerable depth. It is not surprising, therefore, that the TOGAF Standard makes grateful use of the material already contained in the BIZBOK Guide.

A notable part of this guide is the definition of Enterprise Architecture. This results in a third definition of Enterprise Architecture within the same TOGAF Standard. The BIZBOK Guide bases its definition on the Federation of Enterprise Architecture Professional Organizations (FEAPO):

Enterprise Architecture represents the holistic planning, analysis, design, and implementation for the development and execution of strategy by applying principles and practices to guide organizations through the integration and interoperation of all other architecture domains.

What is good to see reflected in the above definition is the relationship to organizational strategy (*for the development and execution of strategy*) and thus to business goals and objectives. This link is missing from the TOGAF Standard's own definition.

Definitions are incredibly important and can make all the difference in understanding what things are about (or what is out of scope or specifically not about).

Ask someone to describe an apple. Nine times out of ten, the person will come up with something like *"a round piece of fruit with a stem (and a small leaf) that contains vitamins"*. Some will also add a color to the description.

However, this description could just as easily refer to a cherry or a grape.

Granted, cherries and grapes are also fruits, but they are not apples. To avoid *comparing apples and oranges*, definitions are used. They really matter [9].

The fact that there are multiple definitions of the same topic within a single standard, such as the TOGAF Standard, is an indication of how difficult a topic this is. The field of Enterprise Architecture has proven over the years to be hard to capture in a single, conclusive definition. In addition, the discipline manifests itself in different ways and at different levels, making it difficult to define what it is.

The conclusion is that there is no single, definitive definition of Enterprise Architecture. Therefore, it is particularly important to define the way Enterprise Architecture is practiced in an organization or how it is represented, so that there is no ambiguity within the organization.



Further Reading

An interesting article on the variety of definitions can be found at the URL below:

<https://eawheel.com/blog/2023/04/the-many-definitions-of-enterprise-architecture/>

4.4. Summary

Enterprise Architecture is a difficult topic to define. This has been proven over the years. The fact that there are multiple definitions for the same thing is ample evidence of this. Within the TOGAF Standard alone, three definitions are used:

- A definition based on the ISO/IEC/IEEE 42010:2022 standard
- A second definition from Gartner
- A third definition, taken indirectly from the BIZBOK Guide

For the first definition, the TOGAF Standard adopts the ISO/IEC/IEEE 42010:2022 standard for defining Enterprise Architecture. The framework adds a second definition to the existing one, in order to make it more complete and fitting. What is interesting about the TOGAF Standard's definition of Enterprise Architecture is that a key component is missing. It is about the realization of business goals.

For the second definition - described in the Leader's Guide - the TOGAF Standard refers to Gartner. This organization created a definition of Enterprise Architecture in 2008. This is the one adopted in the TOGAF Standard, which was updated in 2025 with the Technical Corrigendum 1. The risk of adopting a definition that is several years old is that it may now appear outdated. There is a real chance that the definition being referenced has since been updated and revised.

The third definition of Enterprise Architecture is indirectly adopted by the TOGAF Standard from the BIZBOK Guide. Since the TOGAF Standard refers to the BIZBOK Guide for the design and application of the Business Architecture domain, the definition of Enterprise Architecture used there is also adopted.

Three definitions in one standard.

The profession of Enterprise Architecture is difficult to capture in a single definition. Therefore, it is more important to define the way Enterprise Architecture is practiced in an organization so that there is no ambiguity within the organization.